

ules is another revenue-generating area and, hence, an important driving factor in the energy sector. India can become a well-established, low-cost producer and assembler of solar PV cells and modules. According to India Semiconductor Association (ISA), it is likely that grid connected power would emerge as a major segment in India over the next few years. As per ISA, the country currently has an installed manufacturing capacity of solar cells and modules of 400 MW and 700 MW, respectively.

"In India, monocrystalline have been the fastest moving cells with efficient technologies. Most of the rooftop panels are based on monocrystalline cells," explains Rajiv Kulshretha, manager-marketing and sales, Steca-Bergen Solar Products. "Presently, most of the solar cells and modules are being imported. And those who are manufacturing in India are largely exporting to Europe and the US," he adds.

Thin-film is another fast growing market. Thin-film solar cells are rapidly taking market share away from the established crystalline technology, according to iSuppli Corp. iSuppli points out that, globally, thin-film market will double by 2013. The average thin-film solar panel price is expected to decline to \$1.40 in 2010, down 17.6 per cent from \$1.70 in 2009. Moser Baer's PV subsidiary is ready for production of thin-film PV modules at its manufacturing plant in Greater Noida.

Presently, with a tiny domestic market (around 2.5 MW in 2008), the Indian PV industry looks insignificant. But with the approval of the National Solar Mission, the industry is expected to grow fast. "After the cabinet's nod to the National Solar Mission, suddenly we have started getting several inquiries from customers to set up plants for them," informs Grover.

Global glut is temporary

The global glut of solar panels that has overwhelmed the industry for much of 2009 is now nearing its end with strong demand from Germany, the world's largest solar market. In August, the

iSuppli pegged oversupply at 92 per cent for 2009 but it recently revised the average rate to 66 per cent.

According to the new report by iSuppli, "The oversupply problem plaguing the solar panel industry may come to a close. This glut will cause prices to plunge this year. An increase in demand from Germany will help to consume the excess inventory on the market, which should provide some support to the manufacturers."

As the industry adjusts, production is expected to flatten from 2011 through 2013. By the end of 2009, solar panel production will increase by 14.3 per cent to 7.5 GW from 6.5 GW in 2008. But, only 3.9 GW of installations will occur this year. As a result, close to half of all panels will be in excess. An expected acceleration in demand after 2012 will absorb the excess inventory and lead to more production, says the report.

Despite the current situation, most manufacturers are not pulling back on supply in order to maintain their market shares.

When will India reach grid parity?

Notwithstanding this temporary glut in the global scenario, industry experts are bullish about the long-term viability of the Indian PV market. "By 2010, cell and module prices will fall and returns on solar farms will certainly improve. Previously, we were thinking it will be 2015 before we will reach the grid parity, but now it seems it will be sooner than that. Today, the module prices account for 55-65 per cent of the solar system cost. As the module prices go down, the cost of generating solar power will go down too. The moment we go below \$3 per watt, we will start looking at grid parity in certain applications," explains Arya.

Investments and returns

Is investing in this sector sensible in the present scenario, when awareness, demand and manufacturing are all in nascent stages? "While the government is encouraging manufacturing in the solar domain, and despite suitable

demographics, sunshine and power shortages, players are still slow in taking a decision as the initial investment is huge. It may take more than three years to get the return, depending upon the capacity," says Kulshretha. "Unless you can do 1000 MW of production using innovative technology, have really mastered it, and understood the trends of the Indian market, it is not sensible to compete with the foreign players, particularly the Chinese," says Rajesh. "The only way to compete is through higher volume and lower price," he adds.

So where should the investors put in their capital? "Into module and panel production and domestic power generation," Rajesh is prompt in commenting. "If companies invest in manufacturing solar modules and panels made up of assemblies of PV cells, this would give India a potential edge. One can do good business by importing low-cost PV cells in large volumes and make PV panels here," says Cahandel.

"Solar power generation in India can really make the investors see the sunshine days. This is where they can make money in India," says Srivastava. This is not because of the ample amount of sunshine India has but because it has acute power shortage.

In January 2008, the Union Minister of New and Renewable Energy announced feed-in tariffs (FITs) for solar PV projects up to a maximum capacity of 50 MW. Under the scheme, the project developer makes a power purchase agreement with the state utility at the highest existing market rate. The ministry, through the Indian Renewable Energy Development Agency, augments this rate to a maximum of Rs 15 per kWh. The maximum supplement incentive from the ministry is restricted to Rs 12 per kWh. This will be reduced by 5 per cent for projects commissioned from 2010 onwards. The supplement is available for up to 10 years. The announcement saw a competition amongst the investors for the 'gold rush', filing papers to set up solar power projects. But after a year, nothing has taken shape.

Some of the states like Gujarat have